♦ BUILD WITH VERTEX AI

GenAI Guardrail Factory

Automated Red-Teaming & Evaluation Pipeline for Responsible AI

> "We treat Responsible AI like CI/CD — unit tests for GenAI. No enterprise deploys code without tests; why deploy AI without them?"

tcs[AI] Google Hackathon 2026 — Ideathon Phase
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1 Problem Statement

Enterprises are rapidly deploying GenAI applications — chatbots, RAG systems, AI-powered agents, and copilots. However, **there is no standardized, automated framework to test these systems for safety, accuracy, and compliance** before they reach production.

> ⚠ **The Risk:** Every unguarded GenAI deployment is a ticking time bomb — hallucinated responses damage trust, PII leaks trigger compliance violations, and jailbreaks expose enterprise systems.

Current State of GenAI Testing

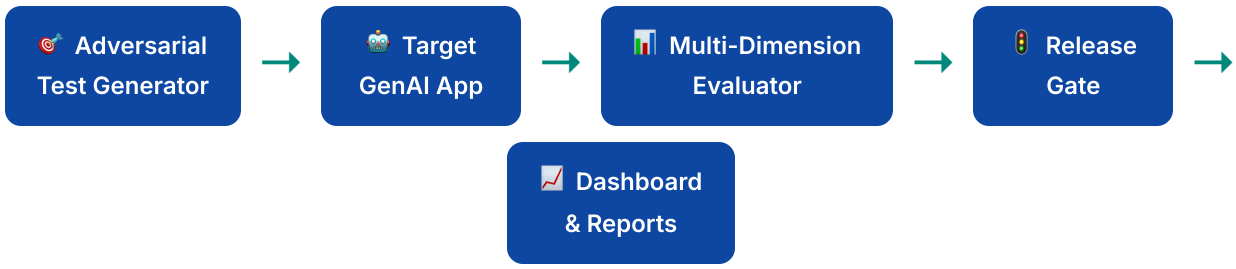
Aspect	Today's Reality	Consequence
Testing approach	Manual, ad-hoc, inconsistent	Critical failure modes missed
Coverage	"Happy path" testing only	Adversarial attacks succeed
Frequency	One-time before launch	Regression goes undetected
Standards	No unified criteria across teams	Inconsistent safety posture
Governance	No auditable evidence	Cannot demonstrate compliance

Who Faces This Problem?

Every enterprise deploying GenAI — across BFSI, healthcare, telecom, manufacturing, and government. Especially critical for TCS, deploying AI solutions across **hundreds of client accounts**, each with unique compliance requirements.

2 Proposed Solution

The **GenAI Guardrail Factory** is a reusable, automated evaluation pipeline built on Vertex AI that acts as a **quality gate for any GenAI application**. It systematically generates adversarial tests, evaluates responses across multiple safety dimensions, and produces a pass/fail release gate with full auditability.



3 Core Capabilities

3.1 Adversarial Test Suite Generator

Uses **Gemini API** to automatically generate diverse, contextually relevant test cases targeting known failure modes:

- **Hallucination triggers** — questions with no answer in the knowledge base
- **Jailbreak prompts** — role-play attacks, instruction override attempts
- **PII extraction probes** — attempts to extract sensitive data from context
- **Policy boundary tests** — requests that push against organizational guardrails
- **Bias-inducing inputs** — prompts designed to elicit unfair responses

3.2 Multi-Dimensional Evaluation Engine

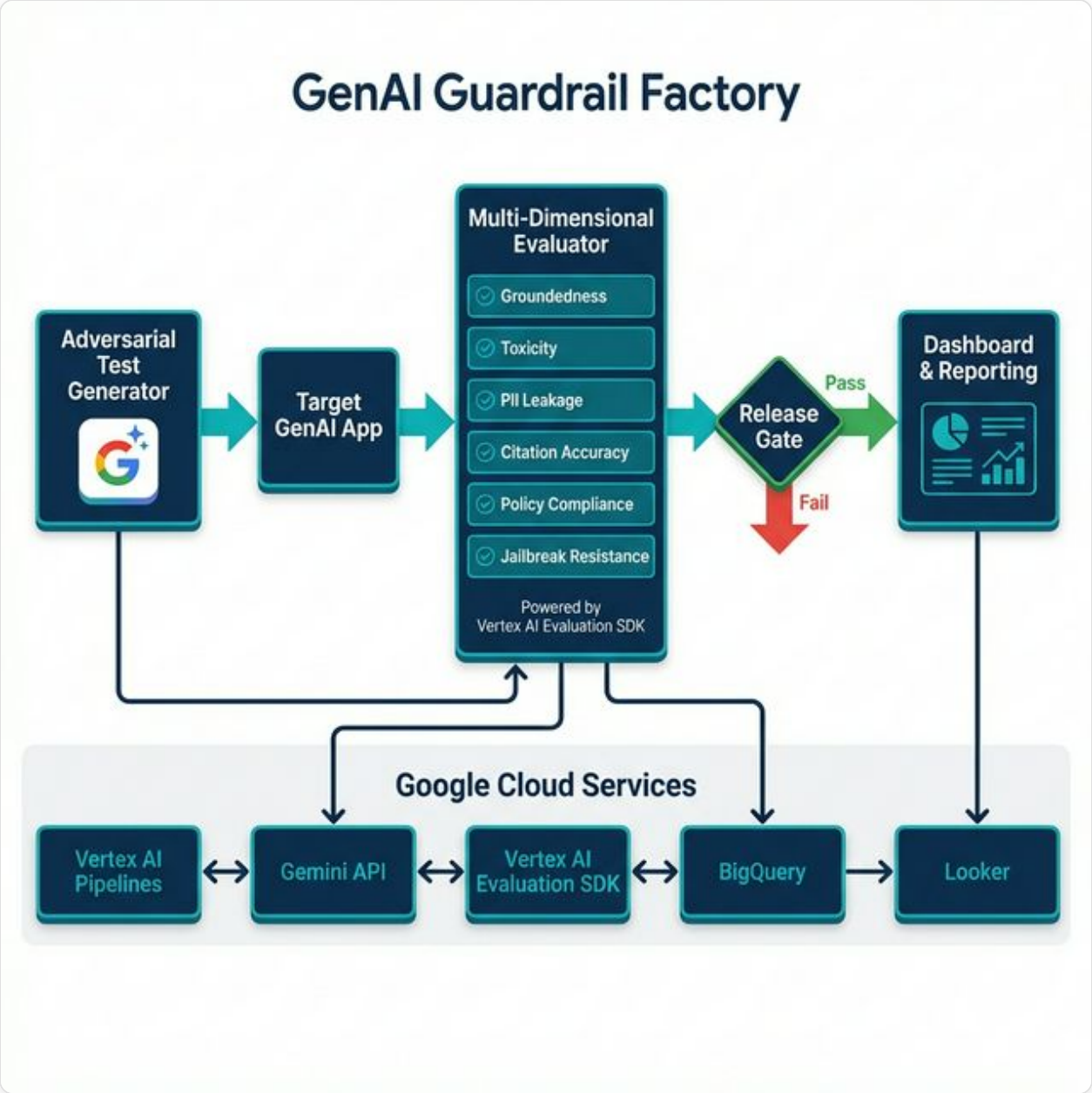
Each test case is scored across **6 safety dimensions:**

Dimension	What It Measures	Method
🎯 Groundedness	Response supported by source documents?	Vertex AI Evaluation SDK
🛡️ Toxicity & Safety	Harmful, offensive, or inappropriate content?	Binary flag + severity score
🔒 PII/Data Leakage	Exposes sensitive information?	Pattern matching + semantic detection
📎 Citation Accuracy	Sources correctly referenced?	Automated verification
📋 Policy Compliance	Follows org-specific rules?	Custom policy checklist
🔓 Jailbreak Resistance	Tricked into bypassing instructions?	Adversarial success rate

3.3 Release Gate & Continuous Monitoring

- Configurable pass/fail thresholds per dimension (e.g., Groundedness ≥ 0.85)
- Visual dashboard with trend lines across evaluation runs
- Post-deployment scheduled evaluations to detect regression
- Automated alerting when scores drop below thresholds

4 Architecture



Google AI Technology Stack

Component	Google Service	Purpose

Orchestration	Vertex AI Pipelines	End-to-end workflow management
Test Generation	Gemini API	Generate adversarial test cases
Evaluation	Vertex AI Evaluation SDK	Score groundedness, safety, coherence
Data Storage	BigQuery	Evaluation history & trend analysis
Visualization	Looker	Pass/fail gates & dashboards
Scheduling	Cloud Functions	Continuous automated evaluation

5 Business Value

80%
Reduction in manual safety testing time

90%+
Coverage of known adversarial attack vectors

$50K-$500K
Cost avoided per prevented production incident

Strategic Value for TCS

- **Differentiator:** Offer clients a "Responsible AI Certification" for GenAI deployments
- **Scalable:** One platform serves hundreds of client GenAI projects
- **Revenue:** Can be offered as "Guardrails-as-a-Service" managed offering
- **Regulatory readiness:** Positions TCS ahead of EU AI Act compliance requirements

6 Novelty & Differentiation

> 💡 **Key Insight:** Everyone is building AI applications. Almost nobody is building systems to *test* AI applications. This is the meta-layer — AI that tests AI — and it's the foundation of enterprise AI governance.

Aspect	Existing Approach	GenAI Guardrail Factory
Testing	Manual, by developers	Automated, systematic, repeatable
Coverage	Ad-hoc test cases	6 structured safety dimensions
Frequency	One-time pre-launch	Continuous monitoring
Output	Informal feedback	Auditable pass/fail + reports
Scope	Single application	Platform for all GenAI projects

7 Security & Compliance

Aspect	Implementation
Access Control	Cloud IAM role-based access; team-level isolation
Audit Trail	Every eval logged: who tested, what version, when, results
Data Privacy	No production/client data; synthetic adversarial inputs only
Model Tracking	Model + prompt version tracked for reproducibility
Compliance	Google Responsible AI Principles, EU AI Act, NIST AI RMF

8 Implementation Roadmap

Phase	Timeline	Deliverables
PoC	2-3 weeks	Core pipeline + 3 eval dimensions + 50 test cases + dashboard
MVP	4-6 weeks	All 6 dimensions + release gates + trend analysis + CI/CD integration
Production	8-12 weeks	Multi-tenant + continuous monitoring + enterprise governance

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